MERN STACK

DAY – 34

3 August 2026

PROJECT: PARKEASY - PARKING MANAGEMENT SYSTEM

AVAILABLE PARKING SLOTS MODULE DEVELOPMENT

• The Available Parking Slots module is developed to allow users to view and select available parking spaces in the ParkEasy parking management system.
• It helps users find suitable parking slots based on their preferred date and time.
• The module prevents booking conflicts by checking already reserved slots during the selected time period.
• The useState() hook is used to manage:

1. Available Slots
2. Search Value
3. Selected Slot
4. Booking Details
5. Booking Form Visibility

• The useEffect() hook is used to automatically fetch available parking slots when the selected date and timing are changed.

• The page displays parking slot information including:

1. Slot Number
2. Availability Status
3. Selected Timing
4. Booking Request Option

• A search feature is provided to help users quickly find specific parking slots.
• Users can select a slot and send a booking request after confirming the required details.

AVAILABLE SLOTS API DEVELOPMENT

• The Available Slots API is created using Express.js to retrieve parking slot information from MongoDB.

• The API checks slot availability based on maintenance status and existing bookings.

API Endpoint:

```
GET /slots
```

• Parking slots are retrieved from the Slot collection using:

```
Slot.find()
```

• Slots marked as Maintenance are removed from the available slot list.

• When date and time details are provided, the API checks existing bookings from the User collection.

• User booking records are verified based on:

1. Booking Date
2. Start Time
3. End Time
4. Booking Status

• Cancelled, Rejected, and Completed bookings are ignored while checking slot availability.

• The API compares booking time intervals and removes already booked slots.

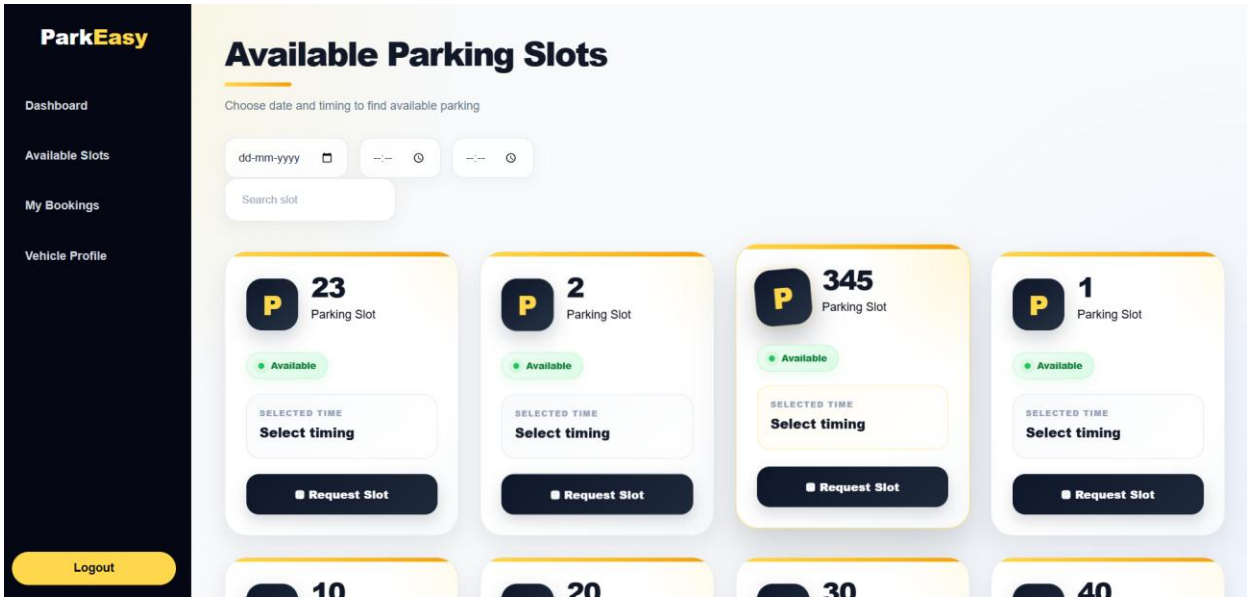
• The API returns only available parking slots to the frontend.

Working

1. The AvailableSlots() component is created to display available parking spaces.
2. The component sends a request to the /slots API using Fetch API.
3. The backend retrieves parking slot details from the Slot collection.

4. The system checks user booking records to identify unavailable slots.
5. The API filters booked slots based on selected date and time.
6. Available slot details are sent back to the frontend.
7. The received slot data is displayed using React state.
8. Users can search, select, and request a parking slot.
9. The booking confirmation form displays selected slot and timing details.
10. After successful booking, the available slots list is updated automatically.

```
app.get("/slots", async(req,res)=>{
try{
const {
date,
startTime,
endTime
}=req.query;
const slots = await Slot.find({
status:{
$ne:"Maintenance"
}
});
if(!date || !startTime || !endTime){
return res.json(slots);
}
const users = await User.find();
let bookedSlots=[];
users.forEach(user=>{
user.bookings.forEach(booking=>{
if(
booking.date===date &&
booking.status!=="Cancelled" &&
booking.status!=="Rejected" &&
booking.status!=="Completed" &&
(
startTime < booking.endTime &&
endTime > booking.startTime
)
){
bookedSlots.push(booking.slot);
}
});
});
const availableSlots = slots.filter(slot=>
!bookedSlots.includes(slot.slotNumber)
);
res.json(availableSlots);
}
catch(err){
res.status(500).json({
message:err.message
});
}
});
```


ParkEasy
Dashboard
Available Slots
My Bookings
Vehicle Profile
Logout
Available Parking Slots
Choose date and timing to find available parking
dd-mm-yyyy
Search slot
23
Parking Slot
Available
SELECTED TIME
Select timing
Request Slot
2
Parking Slot
Available
SELECTED TIME
Select timing
Request Slot
345
Parking Slot
Available
SELECTED TIME
Select timing
Request Slot
1
Parking Slot
Available
SELECTED TIME
Select timing
Request Slot